

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: The hydrosphere: hydrological cycle & unsustainable exploitation - Set 1

Mark scheme

AQA spec 3.2.2.1-3.2.2.3; The physical environment; textbook pp. 114-121

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $13 / 480 = 0.0271$ years (1) • 0.0271×365 (1) • = 10 days (accept 9.9-10) (1)	3
01.2	<i>1 mark for the correct answer.</i> • $1\,340\,000 / 425 = 3153$ years (accept 3100-3200) (1)	1
01.3	<i>1 mark for each point.</i> • It has a very short residence time / very rapid transfer rate (480×10^{15} dm³ per year) (1) • So it moves large volumes of water each year, delivering precipitation to land; small reservoirs with fast transfers respond quickly to change (1)	2
01.4	<i>1 mark for each point.</i> • The ocean is so large that an added or removed volume is a tiny proportion of the total (1) • The same volume is a large proportion of a small reservoir so its level / residence time changes noticeably (1)	2
01.5	<i>1 mark for each point.</i> • Water is constantly moving between reservoirs (evaporation, precipitation, runoff, infiltration, transpiration) (1) • But inflows and outflows for each reservoir balance so the volume in each reservoir stays constant over long periods (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 115-116; AQA spec 3.2.2.1.	

Question	Marking guidance	Marks
02.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Deforestation: less interception so less evaporation from leaves; less transpiration so less water returned to the atmosphere and less precipitation downwind; more infiltration and surface runoff, higher river flows / flooding (2) • Urban development: impermeable concrete / tarmac reduce infiltration and aquifer recharge; runoff is faster and greater, increasing river flooding downstream; drains carry water quickly to rivers (2) • Agriculture: soil compaction by machinery / trampling reduces infiltration and soil moisture, increases runoff; irrigation increases evaporation; loss of soil biota (worms) reduces infiltration and water retention; drainage lowers water tables (2)	6
02.2	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Higher temperatures increase evaporation and change the amount, timing, type (rain vs snow) and location of precipitation (1) • More rapid melting of snow and ice changes river regimes: more flooding after rain, lower flows in dry periods (1) • Changed wind patterns move humid air to different areas (1) • Sea level rise / salt incursion into coastal aquifers (1)	2
02.3	<i>1 mark for each point.</i> • Most water abstracted is cleaned and returned to the river / sea so the main impact is on quality (pollution) not quantity (1) • Effects are noticeable only where large urban centres draw on restricted local supplies (aquifers, small rivers) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 116-117; AQA spec 3.2.2.1.	
03.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(68\,000 - 7\,000) / 68\,000 \times 100$ (1) • = 89.7% (accept 90%) (1)	2
03.2	<i>1 mark for each point.</i> • Inflow from the rivers was reduced (sometimes to zero) so less water entered while evaporation continued (1) • Evaporation removes pure water leaving the dissolved salts behind, so they became concentrated in a smaller volume (1)	2
03.3	<i>1 mark for each point.</i> • Salinity rose above the range of tolerance of the freshwater fish species / osmotic stress; pollution by pesticides, fertilisers and industrial waste concentrated in the lake (1) • Exposed lake bed of salt and pesticide-contaminated dust blown by wind onto farmland and settlements (health problems, soil salinisation) (1) • Loss of wetland habitats and wildlife; local climate more extreme (loss of the moderating effect of the water body) (1) • Loss of livelihoods; conflict between the countries sharing the rivers (1)	4
03.4	<i>1 mark for each point.</i> • Upstream abstraction for irrigation reduces the flow available downstream (1) • Upstream HEP dams hold water in summer for winter generation, reducing summer irrigation water downstream; growing populations increase demand (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117-118; AQA spec 3.2.2.1-3.2.2.2.	
04.1	<i>1 mark for the correct answer.</i> • $(44 - 8) / 50 = 0.72 \text{ m yr}^{-1}$ (1)	1
04.2	<i>1 mark for each point.</i> • Abstraction (for irrigation) exceeds the rate of natural recharge by infiltration / percolation of precipitation (1) • The dynamic equilibrium between inflow and outflow has been upset so the stored volume declines (1)	2
04.3	<i>1 mark for each point.</i> • Porosity: a high proportion of the rock volume is space that can hold water (chalk, limestone, sandstone) (1) • Permeability: the spaces are interconnected and large enough for water to flow through (clay is porous but not permeable) (1) • Impermeable rock (granite, clay) beneath to stop water escaping, and permeable rock / soil above to allow recharge (1)	3
04.4	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Salt water incursion: seawater flows into the aquifer to replace the fresh water, making it unusable for irrigation (osmotic dehydration of crops) (1) • Subsidence: emptied pore spaces no longer support the rock so it compacts, damaging buildings and pipes (eg Mexico City) (1) • Drying up of groundwater-fed rivers, lakes and wetlands as the water table falls below their beds (1) • Ecological impacts: plants needing high water availability die; wetland animals lost; food chain effects (1) • Reduced supplies: abstraction may suddenly have to stop after years of hidden decline (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	
05.1	<i>1 mark for each point.</i> • The water was recharged during or before the last ice age (10 000-20 000 years ago) when the climate was wetter (1) • Current recharge is little or none so any abstraction reduces the stored volume (1) • The large volume means shortages take a long time to appear, but eventually the supply runs out; ancient water is often saline, causing soil salinisation (1)	3
05.2	<i>1 mark for each point.</i> • Two satellites whose orbits are affected by the force of gravity (1) • The mass of water in aquifers below changes gravity, so changes in the orbit / distance between the satellites indicate changes in groundwater mass (1) • Advantage: the whole Earth is surveyed every 30 days, whereas borehole water-table measurements are slow and hard to coordinate over large areas (1)	3
05.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Low-pressure irrigation pipes to reduce leaks (1) • Growing crops that need less water than wheat / maize (1) • Charging consumers by volume used (metering) (1) • Inter-basin transfer of water 450 km from the Yangtze (1)	3

Question	Marking guidance	Marks
05.4	<i>1 mark.</i> • Trees intercept rainfall and transpire large volumes, reducing soil moisture and the water percolating down to the aquifer (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	
06.1	<i>Up to 6 marks in total; award marks in proportion to the completeness of each point.</i> • Habitat change: flooding destroys the previous terrestrial / river habitats but creates a wetland which may be valuable (wetlands are uncommon) (up to 2) • Wildlife barrier: the dam blocks migrating fish (salmon, sturgeon) and prevents recolonisation along the river (up to 2) • Changed river regime downstream: regulated flow removes low-flow periods needed by species such as river turtles nesting on sandbanks and high flows that wash sediment from gravel spawning beds (up to 2) • Sedimentation: sediment settles in the reservoir instead of fertilising the floodplain and building river banks / coastlines downstream, increasing erosion (up to 2) • Microclimate: water's high heat capacity reduces temperature fluctuations; higher wind speeds; more evaporation, humidity, cloud and precipitation downwind (up to 2)	6
06.2	<i>1 mark for each point.</i> • Vegetation / dead organic matter under water decays anaerobically (1) • Releasing methane, a powerful greenhouse gas (1)	2
06.3	<i>1 mark for each point.</i> • After heavy rain surplus water is stored, reducing peak flow / flooding downstream (1) • In dry periods the dam releases water to maintain river levels / supplies (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117; AQA spec 3.2.2.1.	